

Revised Generation Interconnection System Impact Study Report

For Queue Project Q12-0122

Carbon Center 345 kV
March 31, 2013

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q12-0122. The Interconnection Customer proposes a 276.5 MW wind facility interconnecting at the Carbon Center 345 kV substation in the Pennsylvania area. The request was received on April 04, 2012. The study evaluated system intact and single contingency conditions using the regional planning case, including 106 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$6,653,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Sprintbrook / Woodside Substation 138kV line
Contingency: Shawville Sta / Orchard Rd 500kV line
Loading: 327.5% (Rating: 215 MVA)
DFAX: 5.4%
Pre-project loading: 320.6%

2. Facility: Dekalb 500-1 kV TX
Contingency: Richland / Dekalb 500kV line
Loading: 103.3% (Rating: 969 MVA)
DFAX: 7.3%
Pre-project loading: 101.3%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Sprintbrook / Woodside Substation 138kV line	Rebuild line with higher-capacity conductor	\$3,410,000	57% (\$1,960,000)
Dekalb 500-1 kV TX	Replace transformer with higher-rated unit	\$37,472,000	12% (\$4,693,000)
	Total allocated to project		\$6,653,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.